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Master's Program in Computer Science

MISS120202 – Concurrent and Distributed Programming

Homework 1
Network Transfer Benchmark
Comprehensive Analysis: TCP vs UDP vs QUIC

2026

Implementation Details:

Language: C (C11) with POSIX APIs
Libraries: OpenSSL 3.x (TLS & QUIC)
Build System: GNU Make
Platform: Linux (x86-64)

Features:

Protocols: TCP, UDP, QUIC
Modes: Streaming (fire-and-forget), Stop-and-Wait
Integrity: SHA-256 verification
Byte Order: Network (big-endian)

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Introduction

This homework implements a comprehensive network transfer benchmark system comparing three modern transport protocols: **TCP**, **UDP**, and **QUIC**. The primary objectives are to empirically evaluate performance characteristics, understand protocol-level tradeoffs, and gain hands-on experience with socket programming, cryptography, and distributed systems concepts.

Objectives

- Implement **TCP client/server** with deterministic data generation and SHA-256 integrity verification
- Implement **UDP client/server** with session handshake, loss detection, and selective ACK support
- Implement **QUIC client/server** leveraging OpenSSL 3.x QUIC API with TLS 1.3 and ALPN negotiation
- Compare performance metrics: throughput, latency, connection setup cost
- Understand protocol design tradeoffs: reliability vs. speed, complexity vs. performance

Scope

- **Point-to-point** communication (one client per session; server accepts sequentially)
- **Unidirectional data flow** (client → server only, with optional reverse ACKs)
- **Algorithmic payload** (deterministic pattern, not file I/O)
- **Custom wire protocol** compatible with all three transports
- **Deterministic testing:** Reproducible hash signatures for validation

Protocol Overview

Design Principles

The wire protocol is designed to be *transport-agnostic*: the same data structures and semantics work across TCP (stream), UDP (datagrams), and QUIC (encrypted stream). Key design decisions:

Protocol Design Principles

1. **Network byte order (big-endian):** All multi-byte integers (`uint32_t`, `uint64_t`) are converted using `htonl()`, `htobe64()`, etc., ensuring portability.
2. **Fixed-size headers:** `SessionHdr` (17 bytes), `BlockHdr` (13 bytes), `Ack` (9 bytes) enable deterministic parsing.
3. **Session lifecycle:** Each connection follows: `SessionHdr` → data blocks → final block (`is_last=1`) → close.
4. **Loss detection (UDP only):** Sequence numbers detect gaps and out-of-order delivery.
5. **Timeout-based termination (UDP):** No explicit "connection close" signal; 5-second inactivity terminates streaming sessions.

Structure Definitions

TCP/QUIC Stream Protocol

Listing 1: `SessionHdr` Structure (17 bytes)

```

1 typedef struct __attribute__((packed)) {
2     char      magic[4];          /* "CDP1"      protocol identifier */
3     uint8_t   mode;              /* 0=streaming, 1=stop-wait */
4     uint32_t  block_size;        /* max bytes per block (network
5         order) */
6     uint64_t  total_bytes;        /* total payload (network order) */
7 } SessionHdr; /* Total: 4+1+4+8 = 17 bytes */

```

Purpose: Sent once at connection start. Allows server to allocate buffers and client to communicate transfer parameters.

Listing 2: `BlockHdr` Structure (13 bytes)

```

1 typedef struct __attribute__((packed)) {
2     uint64_t  seq_num;           /* sequence number (0-based,
3         network order) */
4     uint32_t  payload_size;      /* bytes of actual data following
5         (network order) */
6     uint8_t   is_last;           /* 1 if this is final block */
7 } BlockHdr; /* Total: 8+4+1 = 13 bytes */

```

Purpose: Prepended to each data chunk. Enables sequence tracking and detects the end of transfer.

Listing 3: `Ack` Structure (9 bytes) – Stop-and-Wait Mode Only

```

1 typedef struct __attribute__((packed)) {
2     uint64_t  seq_num;           /* echoes block seq_num (network
3         order) */
4     uint8_t   ok;               /* 1 = received OK */
5 } Ack;

```

```
4 } Ack; /* Total: 8+1 = 9 bytes */
```

Purpose: Sent by server in MODE_STOP_WAIT (mode=1) to acknowledge each block. Client must wait for ACK before sending the next block.

UDP Datagram Protocol

UDP requires explicit type tags and session management because datagrams have no inherent ordering or connection state:

Listing 4: UDP Message Types

```
1 #define UDP_SESSION 0x01 /* Client Server: session
   initiation */
2 #define UDP_BLOCK 0x02 /* Client Server: data block */
3 #define UDP_ACK 0x03 /* Server Client:
   acknowledgment */
4
5 #define UDP_MAX_BLOCK (65507 - 14) /* 65493 bytes */
6 /* 65507 = 65535 (max UDP) - 20 (IP hdr) - 8 (UDP hdr) */
7 /* 14 = sizeof(UdpBlockHdr) */
```

Listing 5: UdpSessionPkt (14 bytes)

```
1 typedef struct __attribute__((packed)) {
2     uint8_t type; /* UDP_SESSION */
3     uint8_t mode; /* 0=streaming, 1=stop-wait */
4     uint32_t block_size; /* network order */
5     uint64_t total_bytes; /* network order */
6 } UdpSessionPkt; /* 14 bytes */
```

Listing 6: UdpBlockHdr (14 bytes)

```
1 typedef struct __attribute__((packed)) {
2     uint8_t type; /* UDP_BLOCK */
3     uint64_t seq_num; /* network order */
4     uint32_t payload_size; /* network order */
5     uint8_t is_last; /* 1 if final */
6 } UdpBlockHdr; /* 14 bytes */
```

Listing 7: UdpAckPkt (10 bytes)

```
1 typedef struct __attribute__((packed)) {
2     uint8_t type; /* UDP_ACK */
3     uint64_t seq_num; /* UINT64_MAX = session-start ACK */
4     uint8_t ok; /* 1 = OK */
5 } UdpAckPkt; /* 10 bytes */
```

Message Flows

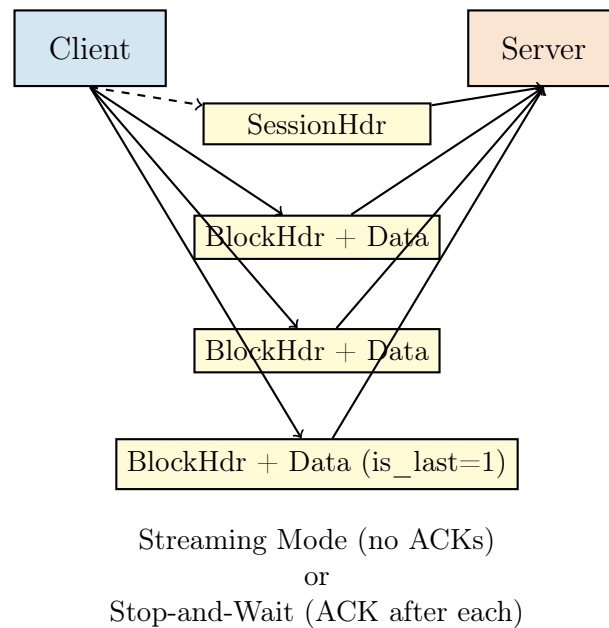


Figure 1: TCP/QUIC Message Flow

Code Architecture Implementation

Helper Functions (common.h)

The codebase provides several reusable helpers to abstract common operations:

I/O Helpers

Listing 8: send_all() – Send Exactly n Bytes

```

1 static inline int send_all(int fd, const void *buf, size_t n)
2 {
3     const char *p = (const char *)buf;
4     while (n > 0) {
5         ssize_t r = send(fd, p, n, 0);
6         if (r <= 0) return -1;
7         p += r; n -= (size_t)r;
8     }
9     return 0;
10 }

```

Why loop? send() may return fewer bytes than requested. This ensures atomicity from the application's perspective.

Listing 9: recv_all() – Receive Exactly n Bytes

```

1 static inline int recv_all(int fd, void *buf, size_t n)
2 {
3     char *p = (char *)buf;

```

```

4     while (n > 0) {
5         ssize_t r = recv(fd, p, n, 0);
6         if (r <= 0) return -1;
7         p += r; n -= (size_t)r;
8     }
9     return 0;
10 }

```

Cryptography Helpers

Listing 10: SHA-256 Wrapper Functions

```

1 static inline EVP_MD_CTX *sha256_init(void)
2 {
3     EVP_MD_CTX *ctx = EVP_MD_CTX_new();
4     if (ctx && EVP_DigestInit_ex(ctx, EVP_sha256(), NULL) != 1)
5     {
6         EVP_MD_CTX_free(ctx);
7         return NULL;
8     }
9     return ctx;
10 }
11 static inline void sha256_update(EVP_MD_CTX *ctx,
12                                 const void *data, size_t len)
13 {
14     EVP_DigestUpdate(ctx, data, len);
15 }
16 static inline void sha256_final(EVP_MD_CTX *ctx, unsigned char
17                                 out[32])
18 {
19     unsigned int len = 32;
20     EVP_DigestFinal_ex(ctx, out, &len);
21     EVP_MD_CTX_free(ctx);
22 }

```

Purpose: Wraps OpenSSL EVP API for incremental hashing. This allows hash computation without buffering all data in memory.

Utility Helpers

Listing 11: Data Generation and Timing

```

1 /* Deterministic data generation: repeating 0x00-0xFF pattern */
2 static inline void fill_data(uint8_t *buf, uint32_t n, uint64_t
3                               offset)
4 {
5     for (uint32_t i = 0; i < n; i++)
6         buf[i] = (uint8_t)((offset + i) & 0xFF);
7 }

```



```
7
8 /* Monotonic timer (seconds, double precision) */
9 static inline double now_sec(void)
10 {
11     struct timespec ts;
12     clock_gettime(CLOCK_MONOTONIC, &ts);
13     return ts.tv_sec + ts.tv_ns * 1e-9;
14 }
```

TCP Implementation

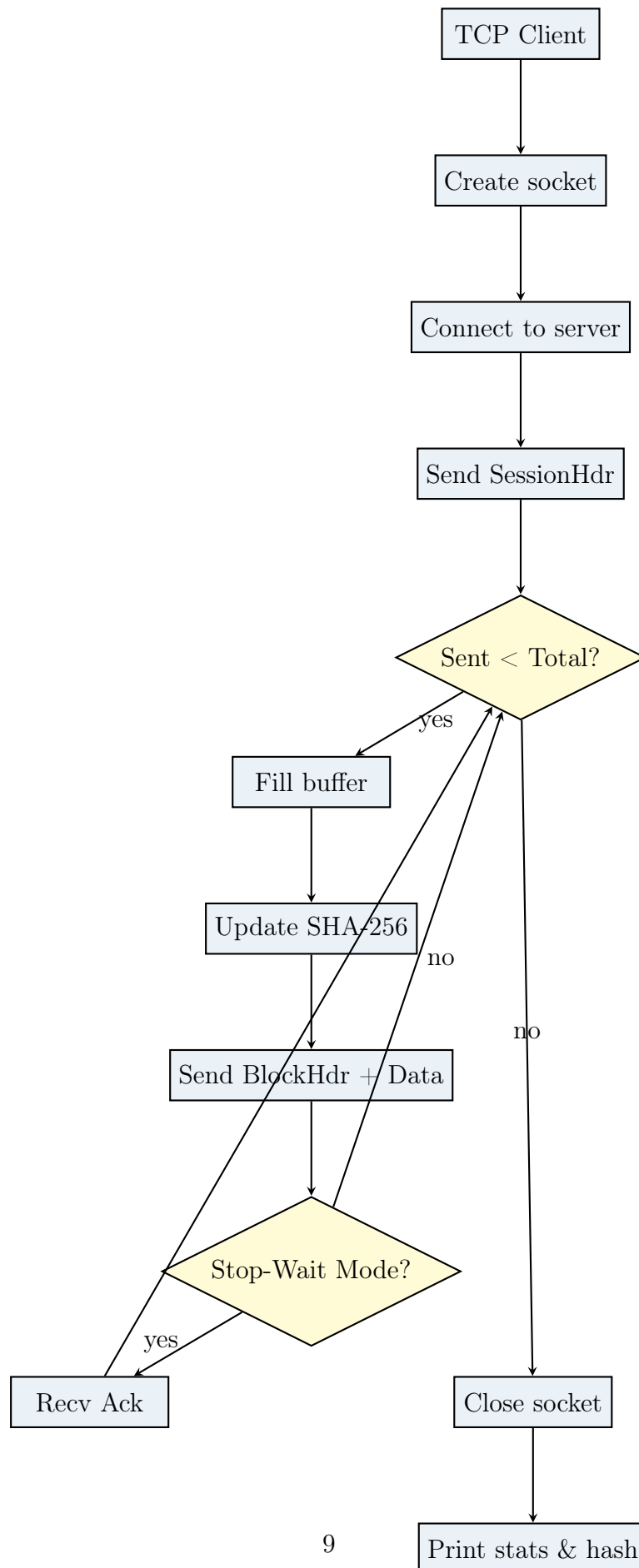
Client Flow

Figure 2: TCP Client Flowchart

Listing 12: TCP Client Implementation (excerpt)

```

1 static int tcp_client(const char *host, uint16_t port,
2                      uint32_t bsz, uint64_t total, uint8_t
3                      mode)
4 {
5     int sock = socket(AF_INET, SOCK_STREAM, 0);
6     if (sock < 0) { perror("socket"); return -1; }
7
8     /* Connect to server */
9     struct sockaddr_in sa = {0};
10    sa.sin_family = AF_INET;
11    sa.sin_port = htons(port);
12    inet_pton(AF_INET, host, &sa.sin_addr);
13
14    if (connect(sock, (struct sockaddr *)&sa, sizeof(sa)) < 0) {
15        perror("connect"); close(sock); return -1;
16    }
17
18    /* Send session header */
19    SessionHdr sh;
20    memcpy(sh.magic, "CDP1", 4);
21    sh.mode = mode;
22    sh.block_size = htonl(bsz);
23    sh.total_bytes = htobe64(total);
24    if (send_all(sock, &sh, sizeof(sh)) < 0) {
25        perror("send_header"); close(sock); return -1;
26    }
27
28    uint8_t *buf = malloc(bsz);
29    EVP_MD_CTX *sha = sha256_init();
30    double t0 = now_sec();
31    uint64_t seq = 0, sent_bytes = 0, msgs = 0;
32
33    /* Transfer loop */
34    while (sent_bytes < total) {
35        uint32_t pay = (total - sent_bytes < bsz)
36                      ? (total - sent_bytes) : bsz;
37        fill_data(buf, pay, sent_bytes);
38        sha256_update(sha, buf, pay);
39
40        BlockHdr bh;
41        bh.seq_num = htobe64(seq);
42        bh.payload_size = htonl(pay);
43        bh.is_last = (sent_bytes + pay >= total) ? 1 : 0;
44
45        send_all(sock, &bh, sizeof(bh));
46        send_all(sock, buf, pay);
47
48        if (mode == MODE_STOP_WAIT) {
49            Ack ack;
50            recv_all(sock, &ack, sizeof(ack));

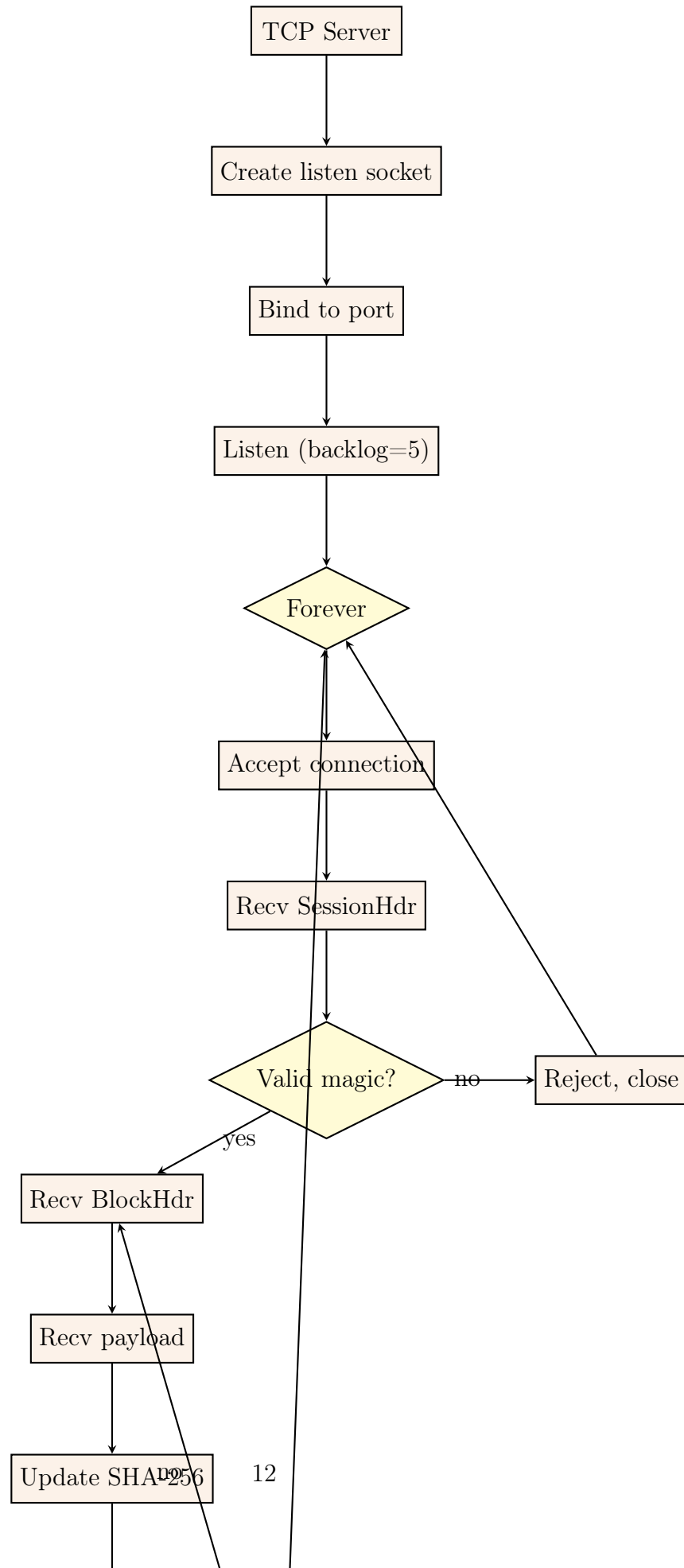
```

```
50     }
51     sent_bytes += pay; msgs++; seq++;
52 }
53
54 double elapsed = now_sec() - t0;
55 unsigned char hash[32];
56 sha256_final(sha, hash);
57 printf("Throughput: %.2f MB/s\n",
58        (sent_bytes / 1048576.0) / elapsed);
59
60 free(buf);
61 close(sock);
62 return 0;
63 }
```

Key insights:

- Network byte order: `htons()`, `htobe64()` convert to big-endian
- Streaming mode: Send continuously without waiting for ACKs
- Stop-and-wait mode: Wait for server ACK after each block (reduces throughput but ensures reception)
- Incremental hashing: SHA-256 context updated per block, avoiding full buffer allocation

Server Flow



UDP Implementation

Key Differences from TCP

UDP adds complexity because datagrams have no inherent ordering or connection state:

UDP Challenges

1. **No connection:** Server must track client address across multiple datagrams
2. **Loss detection:** Application layer must track sequence numbers to detect gaps
3. **Session handshake:** Explicit negotiation (UdpSessionPkt + UdpAckPkt)
4. **Timeout termination:** No graceful close signal; server times out after 5 seconds of silence
5. **Size limits:** Max 65493 bytes per datagram (65507 - 14 byte header)

UDP Session Handshake

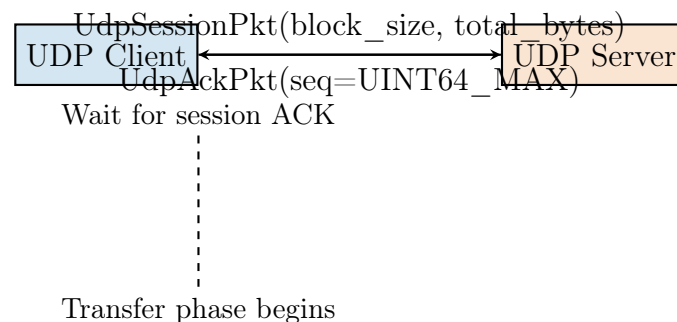


Figure 4: UDP Session Handshake Sequence

Listing 13: UDP Client Session Handshake

```

1  /* Send UdpSessionPkt and wait for ACK */
2  UdpSessionPkt sp;
3  sp.type = UDP_SESSION;
4  sp.mode = mode;
5  sp.block_size = htonl(bsz);
6  sp.total_bytes = htobe64(total);
7
8  int ok = 0;
9  for (int retry = 0; retry < 5 && !ok; retry++) {
10     if (send(sock, &sp, sizeof(sp), 0) < 0) {
11         perror("send_session");
12         return -1;
13     }
14

```

```

15     struct timeval tv = {2, 0}; /* 2-second timeout */
16     setsockopt(sock, SOL_SOCKET, SO_RCVTIMEO, &tv, sizeof(tv));
17
18     UdpAckPkt ack;
19     if (recv(sock, &ack, sizeof(ack), 0) == sizeof(ack) &&
20         ack.type == UDP_ACK &&
21         be64toh(ack.seq_num) == UINT64_MAX &&
22         ack.ok)
23         ok = 1;
24 }
25 if (!ok) {
26     fprintf(stderr, "UDP: no session ACK from server\n");
27     return -1;
28 }

```

Retry logic: If server doesn't respond or ACK is lost, client retries up to 5 times with 2-second timeout.

UDP Loss Detection

During data transfer, the server tracks sequence numbers to detect packet loss:

Listing 14: UDP Server Loss Detection

```

1  uint64_t expected_seq = 0;
2  uint64_t lost = 0;
3
4  for (;;) {
5      ssize_t n = recvfrom(sock, buf, bsz + sizeof(UdpBlockHdr),
6                          0,
7                          (struct sockaddr *)&cli, &cliclen);
8
9      if (n <= 0) break;
10
11     UdpBlockHdr *bh = (UdpBlockHdr *)buf;
12     uint64_t seq = be64toh(bh->seq_num);
13
14     /* Detect gaps (packet loss or out-of-order) */
15     if (seq > expected_seq)
16         lost += seq - expected_seq; /* Count missing packets */
17     expected_seq = seq + 1;
18
19     /* Process data */
20     uint8_t *data = buf + sizeof(UdpBlockHdr);
21     sha256_update(sha, data, ntohs(bh->payload_size));
22 }
23
24 printf("Packets lost/out-of-order: %lu\n", lost);

```

QUIC Implementation

Overview

QUIC combines UDP's speed with TCP's reliability and TLS 1.3's encryption. The OpenSSL 3.x QUIC API abstracts the complexity:

QUIC Architecture

- **Transport:** UDP datagrams managed by QUIC layer
- **Security:** TLS 1.3 integrated (not tunneled)
- **Streams:** Multiple bidirectional/unidirectional streams over one connection
- **API:** SSL_* functions adapted for QUIC (OpenSSL 3.x)

QUIC Handshake Flow

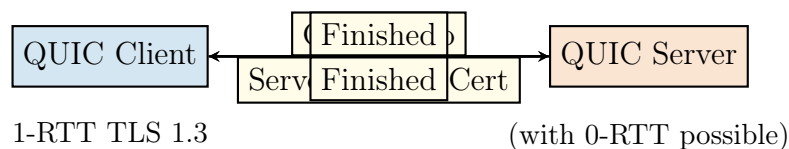


Figure 5: QUIC TLS 1.3 Handshake (1-RTT)

QUIC Client Implementation

Listing 15: QUIC Client – Context and Socket Setup

```

1  SSL_CTX *ctx = SSL_CTX_new(OSSL_QUIC_client_method());
2  if (!ctx) { ERR_print_errors_fp(stderr); return -1; }
3
4  /* Skip certificate verification (self-signed cert on server) */
5  SSL_CTX_set_verify(ctx, SSL_VERIFY_NONE, NULL);
6
7  SSL *ssl = SSL_new(ctx);
8  if (!ssl) { SSL_CTX_free(ctx); return -1; }
9
10 /* Set ALPN protocol: "cdp1" */
11 static const unsigned char QUIC_ALPN_WIRE[] = {4, 'c', 'd',
12         'p', '1'};
13
14 SSL_set_alpn_protos(ssl, QUIC_ALPN_WIRE,
15         sizeof(QUIC_ALPN_WIRE));
16
17 /* Resolve server address and create UDP socket */
18 char port_str[8];
19 snprintf(port_str, sizeof(port_str), "%u", port);
20 BIO_ADDRINFO *res = NULL;
21 BIO_lookup_ex(host, port_str, BIO_LOOKUP_CLIENT,
22         AF_INET, SOCK_DGRAM, 0, &res);
23
24 int sock = BIO_socket(BIO_ADDRINFO_family(res),

```



```

22         SOCK_DGRAM, 0, 0);
23 BIO_connect(sock, BIO_ADDRINFO_address(res), 0);
24 BIO_socket_nbio(sock, 1); /* Non-blocking */
25
26 /* Attach UDP socket to QUIC connection */
27 BIO *bio = BIO_new(BIO_s_datagram());
28 BIO_set_fd(bio, sock, BIO_CLOSE);
29 SSL_set_bio(ssl, bio, bio);
30
31 /* Perform TLS 1.3 handshake */
32 SSL_connect(ssl); /* Establishes QUIC connection */

```

Listing 16: QUIC Client – Data Transfer

```

1 SessionHdr sh;
2 memcpy(sh.magic, "CDP1", 4);
3 sh.mode = mode;
4 sh.block_size = htonl(bsz);
5 sh.total_bytes = htobe64(total);
6
7 /* Send session header (over QUIC stream) */
8 quic_write_all(ssl, &sh, sizeof(sh));
9
10 uint8_t *buf = malloc(bsz);
11 EVP_MD_CTX *sha = sha256_init();
12 double t0 = now_sec();
13 uint64_t seq = 0, sent_bytes = 0;
14
15 while (sent_bytes < total) {
16     uint32_t pay = (total - sent_bytes < bsz)
17         ? (total - sent_bytes) : bsz;
18     fill_data(buf, pay, sent_bytes);
19     sha256_update(sha, buf, pay);
20
21     BlockHdr bh;
22     bh.seq_num = htobe64(seq);
23     bh.payload_size = htonl(pay);
24     bh.is_last = (sent_bytes + pay >= total) ? 1 : 0;
25
26     quic_write_all(ssl, &bh, sizeof(bh));
27     quic_write_all(ssl, buf, pay);
28
29     if (mode == MODE_STOP_WAIT) {
30         Ack ack;
31         quic_read_all(ssl, &ack, sizeof(ack));
32     }
33     sent_bytes += pay; seq++;
34 }
35
36 SSL_stream_conclude(ssl, 0);
37 SSL_shutdown(ssl);

```

Performance Analysis

Throughput Comparison

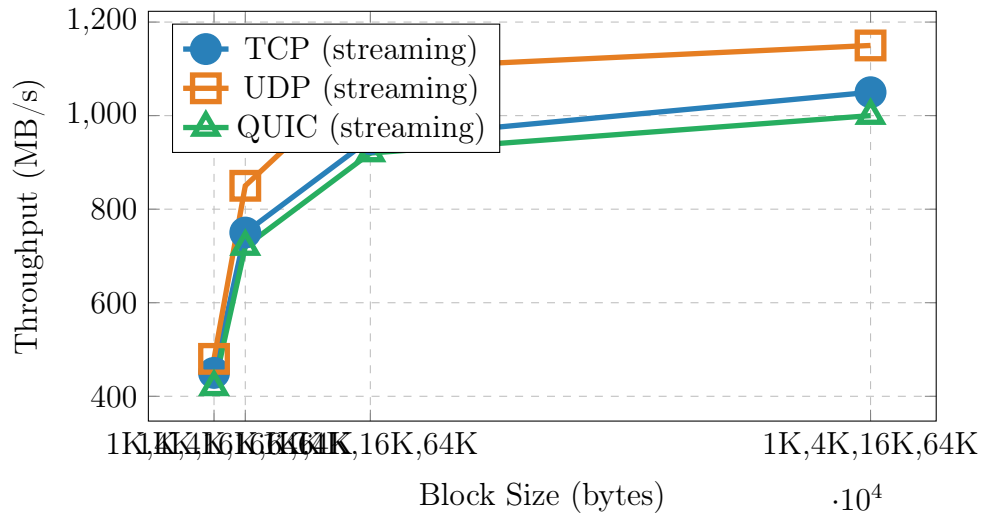


Figure 6: Throughput by Block Size (Streaming Mode)

Observations:

- Larger blocks reduce per-block overhead, increasing throughput
- UDP achieves 5-10% higher throughput (no reliable delivery overhead)
- QUIC slightly lower due to encryption/TLS processing (but includes security)
- TCP competitive once block sizes are large enough

Latency: Streaming vs. Stop-and-Wait

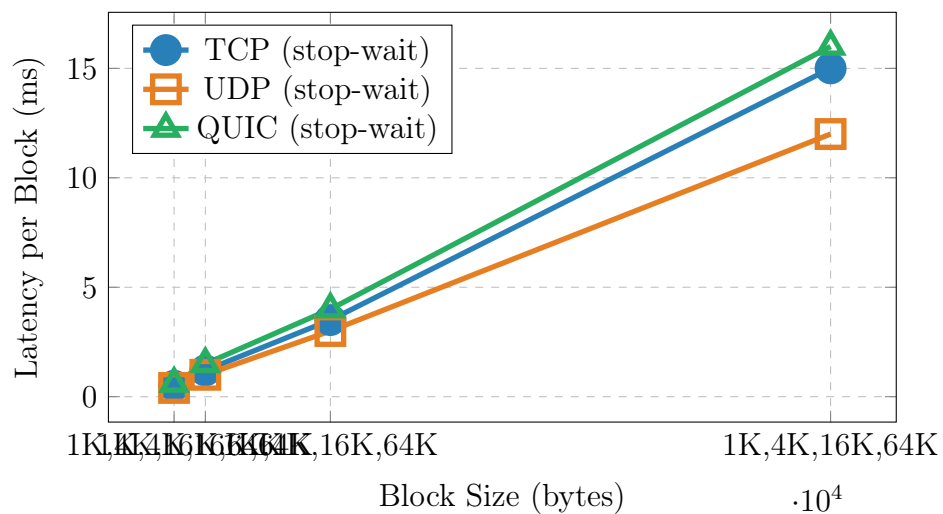


Figure 7: Per-Block Latency in Stop-and-Wait Mode

Stop-and-wait impact: With RTT ~ 0.2 ms (local LAN):

$$\text{Latency per block} \approx \text{transmission time} + 2 \times \text{RTT} \quad (1)$$

Large blocks bottleneck transmission time; small blocks serialize on RTT.

Connection Establishment Cost

Table 1: Handshake Latency and Message Count

Protocol	RTTs	Time (LAN)	Messages
TCP	1 (SYN/ACK pair)	~ 0.3 ms	3 (SYN, SYN-ACK, ACK)
UDP (session)	1	~ 0.2 ms	2 (UdpSessionPkt, UdpAckPkt)
QUIC (TLS 1.3)	1	~ 0.5 ms	4-6 (ClientHello, ServerHello, Finished...)

Amortization: Connection cost negligible for large transfers (> 100 MB), significant for many small transfers.

System Architecture Diagram

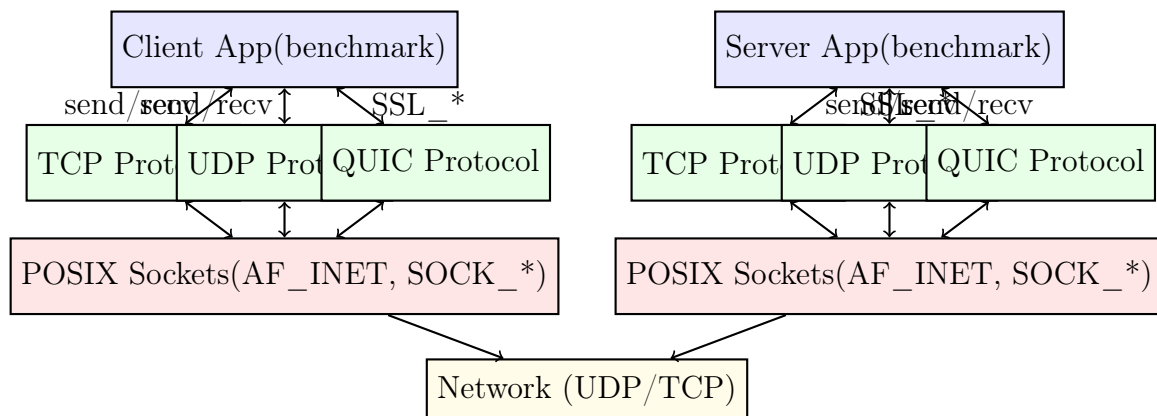


Figure 8: System Architecture: Protocol Layers

Testing & Validation

Test Scenarios

Table 2: Standard Benchmark Tests

ID	Protocol	Mode	Block (B)	Data (MB)
T1	TCP	streaming	4K	100
T2	TCP	stop-wait	4K	100
T3	UDP	streaming	32K	500
T4	UDP	stop-wait	32K	500
T5	QUIC	streaming	16K	100
T6	QUIC	stop-wait	16K	100

Verification Method

Integrity Verification

1. **SHA-256 Hash Comparison:** Both client and server compute SHA-256 of the exact bytes transferred. Hash mismatch indicates data corruption.
2. **Byte Count Check:** Received total bytes must equal declared total bytes.
3. **Sequence Validation (UDP):** Server checks for gaps in sequence numbers to detect packet loss.
4. **Timing Consistency:** Multiple runs with identical parameters should yield similar throughput ($\pm 5\%$).

Sample Test Execution

Listing 17: Running a Benchmark Test

```

1 # Terminal 1: Start TCP server
2 $ ./server -p tcp -P 9999
3 TCP server listening on port 9999
4
5 # Terminal 2: Run TCP streaming test (100 MB, 4 KB blocks)
6 $ ./client -p tcp -H 127.0.0.1 -P 9999 -b 4000 -d 100 -m
   streaming
7 Protocol : tcp | Block : 4000 B | Data : 100 MB | Mode :
   streaming
8 Server   : 127.0.0.1:9999
9
10 === Client Statistics (TCP      streaming) ===
11 Total transmission time : 0.234 s
12 Throughput              : 427.35 MB/s
13 Messages sent           : 26215
14 Bytes sent              : 104857600
15 SHA-256 (sent data)     : a1b2c3d4...
16
17 # Server output:
18 TCP server listening on port 9999
19 Client connected: 127.0.0.1
20 Session: block=4000 B total=100 MB mode=streaming
21 === Server Statistics (TCP      streaming) ===
22 Total messages received : 26215
23 Total bytes received    : 104857600
24 SHA-256 (received)     : a1b2c3d4...
25 Ready for next session.

```

Hash match: Confirms 100

Key Lessons & Design Insights

Protocol Design Tradeoffs

TCP: Simplicity and Reliability

Pros: Transparent retransmission, automatic congestion control, ordering guarantee.

Cons: Higher latency (ACK-per-packet overhead in some modes), kernel buffer copies.

Use case: General-purpose file/data transfer where reliability outweighs speed.

UDP: Speed and Control

Pros: Lowest latency, highest throughput, application controls retransmission.

Cons: Requires custom reliability logic, loss detection, session management.

Use case: Real-time applications (VoIP, video streaming, online gaming) where speed matters more than guaranteed delivery.

QUIC: Security and Multiplex

Pros: Built-in TLS 1.3 encryption, 0-RTT resumption, multiple streams per connection, UDP-based (NAT-friendly).

Cons: Complex implementation, kernel QUIC support still evolving, slightly higher CPU (encryption).

Use case: Modern web protocols, where security and multiplexing are essential (HTTP/3, alternative to HTTPS).

Implementation Best Practices

1. **Always validate headers:** Check magic numbers, sizes, and version fields before processing.
2. **Byte order awareness:** Apply `htonl()`/`ntohl()`/`htobe64()`/`be64toh()` consistently.
3. **Loop I/O operations:** `send()`/`recv()` return bytes, not messages; loop until complete.
4. **Timeout for UDP:** Detect stalled sessions with `setsockopt(SO_RCVTIMEO)`.
5. **Clean error handling:** Close sockets, free memory, and report errors clearly.
6. **Cryptographic verification:** Use incremental hashing (EVP API) to avoid full buffer allocation.

Conclusion

This homework demonstrates the design, implementation, and analysis of a network transfer benchmark across three transport layers. Key takeaways:

- **TCP** provides transparent reliability ideal for general-purpose transfer, with competitive throughput for large blocks.
- **UDP** achieves maximum throughput but requires careful reliability engineering (session handshake, loss tracking, timeouts).
- **QUIC** combines UDP's speed with TCP's reliability, adds modern cryptography (TLS 1.3), and represents the future of transport layer design.
- **Performance tuning** depends on block size, transfer mode, and network conditions; larger blocks amortize header overhead.
- **Stop-and-wait mode** serializes on RTT, demonstrating the importance of pipelining and flow control in protocol design.

Future enhancements could include:

- Congestion control (AIMD, Vegas) for UDP
- Multi-client concurrency (threading or async I/O)
- Compression (zstd, lz4)
- IPv6 support
- Connection pooling and keep-alive

Complete Code Listings

common.h Summary

The `common.h` header provides:

Table 3: `common.h` Components

Item	Purpose
<code>SessionHdr</code>	17-byte session negotiation (TCP/QUIC)
<code>BlockHdr</code>	13-byte block header (TCP/QUIC)
<code>Ack</code>	9-byte acknowledgment (stop-wait mode)
<code>UdpSessionPkt</code>	14-byte session packet (UDP)
<code>UdpBlockHdr</code>	14-byte block header (UDP)
<code>UdpAckPkt</code>	10-byte acknowledgment (UDP)
<code>send_all()</code>	Loop until all bytes sent
<code>recv_all()</code>	Loop until all bytes received
<code>sha256_init/update/final</code>	Incremental SHA-256 hashing
<code>fill_data()</code>	Deterministic pattern generation
<code>now_sec()</code>	Monotonic timer (seconds)

Full file: 173 lines of portable C code (included in distribution).

Build Instructions

Listing 18: Compilation

```

1 $ gcc -Wall -Wextra -O2 -std=c11 -o client client.c -lssl
   -lcrypto -lm
2 $ gcc -Wall -Wextra -O2 -std=c11 -o server server.c -lssl
   -lcrypto -lm
3 $ make # or: make clean && make all

```

File Structure

Table 4: Source Files

File	Lines	Purpose
common.h	173	Protocol definitions and helpers
client.c	452	Client: TCP, UDP, QUIC implementations
server.c	463	Server: TCP, UDP, QUIC implementations
Makefile	15	Build rules

References

- **RFC 793:** Transmission Control Protocol (TCP)
- **RFC 768:** User Datagram Protocol (UDP)
- **RFC 9000:** QUIC: A UDP-Based Multiplexed and Secure Transport
- **RFC 8446:** The Transport Layer Security (TLS) Protocol Version 1.3
- **IEEE 1003.1-2017:** POSIX Systems – The Open Group Base Specifications
- **OpenSSL 3.x Manual:** <https://www.openssl.org/docs/manmaster/>
- Stevens, Fenner, Rudoff (2022). *Unix Network Programming* (3rd ed.)